Boolean Algebra Intermediate Division

1. **00-01 C3 Boolean Algebra**

   Simplify the following Boolean expression: $A(A+BC)+B(B+AC)$

2. **01-02 C2 Boolean Algebra**

   Simplify completely: $(\overline{X}+Y)(\overline{X}+\overline{Y})$

3. **01-02 C2 Boolean Algebra**

   List all ordered triples (A,B,C) that make the following expression FALSE:

   $$\overline{AB}+A(\overline{B+C})$$

4. **01-02 C3 Boolean Algebra** **SKIP**

   Simplify so that only AND and NOT symbols are used:

5. **02-03 C2 Boolean Algebra**

   Simplify: (NOT A OR A AND NOT B) AND (NOT A AND B)

6. **02-03 C2 Boolean Algebra**

   Find all ordered triples that make the following expression TRUE:

   NOT A AND C OR B AND (NOT (A OR C))

7. **02-03 C3 Boolean Algebra** **SKIP**

   Simplify the following expression:

8. **03-04 C2 Boolean Algebra**

   Find all ordered pairs that make the following expression TRUE.

   $$\overline{AB+\overline{A}\,\overline{B}}$$

9. 03-04 C2 Boolean Algebra

Simplify the following expression: $B(\overline{A}+B)+A(\overline{B}+C)+\overline{C}(A+B)$

10. 03-04 C3 Boolean Algebra

Simplify the following Boolean expression: $(A+\overline{B})(\overline{AB})\ (\overline{\overline{A}+B})$

11. 04-05 C2 Boolean Algebra

Simplify: $\overrightarrow{(A+B)}\ (A\overline{B}+B\overline{C}\ )$

12. 04-05 C2 Boolean Algebra

How many ordered triples make the following expression FALSE?

$$\overline{A}(B+\overline{C})+A\overline{B}C$$

13. 04-05 C3 Boolean Algebra

Simplify the following Boolean expression so that the answer uses NOT operator and only one other type of operator:

$$\overline{A+\overline{B}}\quad\overline{C\overline{A}(B+C)}$$

14. 05-06 C2 Boolean Algebra

Simplify: $(\overline{A+\overline{B}})(A\overline{B}+C)(B+\overline{C})$

15. 05-06 C2 Boolean Algebra

How many ordered pairs make the following expression TRUE?

$$(\overline{A}+\overline{B})\overline{C}+\overline{A}(B\overline{C}+A)+(\overline{C+A})(\overline{A}B)$$

16. 05-06 C3 Boolean Algebra

Simplify the expression until only AND and NOT gates are used.

$$\overline{\overline{\overline{A} + A\overline{B}}(B + C)C}$$

17. 06-07 C2 Boolean Algebra

Simplify completely: $\overline{A}(B + \overline{A}) + \overline{A}(\overline{B} + A)$

18. 06-07 C2 Boolean Algebra

List all the ordered triples that make the following expression TRUE:

$$A(\overline{B} + C) + \overline{A}\,\overline{B}(A + \overline{B})C$$

19. 06-07 C3 Boolean Algebra

How many ordered triples make the following expression TRUE?

$$AB\overline{C} + A\overline{B}\,\overline{C} + \overline{A}BC + \overline{A}\,\overline{B}C$$

20. 07-08 C2 Boolean Algebra

Simplify completely: $\overline{A}(AB + C) + B(\overline{A}C + \overline{B})$

21. 07-08 C2 Boolean Algebra

What ordered triples make the following expression TRUE?

$$\overline{A}(A + B\overline{C}) + B\overline{C}(A + \overline{B})$$

22. 07-08 C3 Boolean Algebra

How many ordered triples make the following expression TRUE?

$$(A \oplus B)(A + BC)$$

23. 08-09 C2 Boolean Algebra

Simplify completely: $AB(\overline{\overline{B} + C}) + \overline{C}(\overline{AC + \overline{B}})$

24. 08-09 C2 Boolean Algebra

List all ordered triples that make the following expression TRUE?

$$A(\overline{A\overline{B}C + \overline{A}B\overline{C}})(\overline{A(\overline{B}C + A\overline{C})})$$

25. 08-09 C3 Boolean Algebra

Simplify the following expression: $(A + BC)(\overline{\overline{A}B + \overline{C}})$

26. 09-10 C2 Boolean Algebra

Simplify the following Boolean expression: $(A\overline{B} + B\overline{C})(\overline{A}C + B)$

27. 09-10 C2 Boolean Algebra

Which ordered triple(s) make the following TRUE? $\overline{\overline{A(\overline{B} + C)} + \overline{A}B + C}$

28. 09-10 C3 Boolean Algebra

Simplify the following expression: $\overline{(AB + \overline{C}) + (A(\overline{B} + C))}$

29. 10-11 C2 Boolean Algebra

List the ordered triples that make the following Boolean expression FALSE.

$$A\overline{B}(B + C) + \overline{A(\overline{B + C})(B + \overline{C})}$$

30. 10-11 C2 Boolean Algebra

Simplify the following Boolean expression: $A(\overline{A\overline{B} + BC})$

31. 10-11 C3 Boolean Algebra

How many ordered triples make the following Boolean expression FALSE?

$$A(\overline{A}B + C) + \overline{\overline{B}(A + BC)} + C(\overline{A}B + \overline{B}C)$$

32. 11-12 C3 Boolean Algebra

Simplify the following Boolean expression: $\overline{AB} + B(\overline{A+C})$

33. 11-12 C3 Boolean Algebra

How many ordered triples make the following Boolean expression TRUE?

$$\overline{A}(B+C) + (\overline{A+C})(\overline{BC})$$

34. 12-13 C3 Boolean Algebra

Simplify the following Boolean expression: $(\overline{A}B + \overline{B})(A+B)(A\overline{B} + \overline{A})$

35. 12-13 C3 Boolean Algebra

How many ordered triples make this Boolean expression TRUE?

$$(A + \overline{B}C)(BC + \overline{A}) + A(B + \overline{C})(\overline{A} + \overline{B}\,\overline{C})$$

36. 13-14 C3 Boolean Algebra

Which ordered triple(s) make the following Boolean expression TRUE?

$$(A\overline{B} + B\overline{C})(\overline{A}B\overline{C} + A\overline{B}C)$$

37. 13-14 C3 Boolean Algebra

How many ordered triples make this Boolean expression TRUE?

$$A\overline{B} + B(C + \overline{A}) + B\overline{C} + AB\overline{C}$$

38. 14-15 C3 Boolean Algebra

Simplify the following Boolean expression:

$$(A\overline{B} + AC + B\overline{C})(B\overline{C} + \overline{A}B + \overline{A}C)$$

39. 14-15 C3 Boolean Algebra

How many ordered triples make this Boolean expression FALSE?

$$\overline{AB}(\overline{\overline{A}+B+C})(\overline{A+B\overline{C}})$$

40. 15-16 C3 Boolean Algebra

Which ordered triple(s) make the following Boolean expression TRUE?

$$\overline{A}(B+A\overline{C})+B(C+\overline{A}B)$$

41. 15-16 C3 Boolean Algebra

How many ordered triples make the following Boolean expression TRUE?

$$\overline{A}(\overline{B+A\overline{C}})+\overline{BC}(\overline{\overline{A}+C})$$